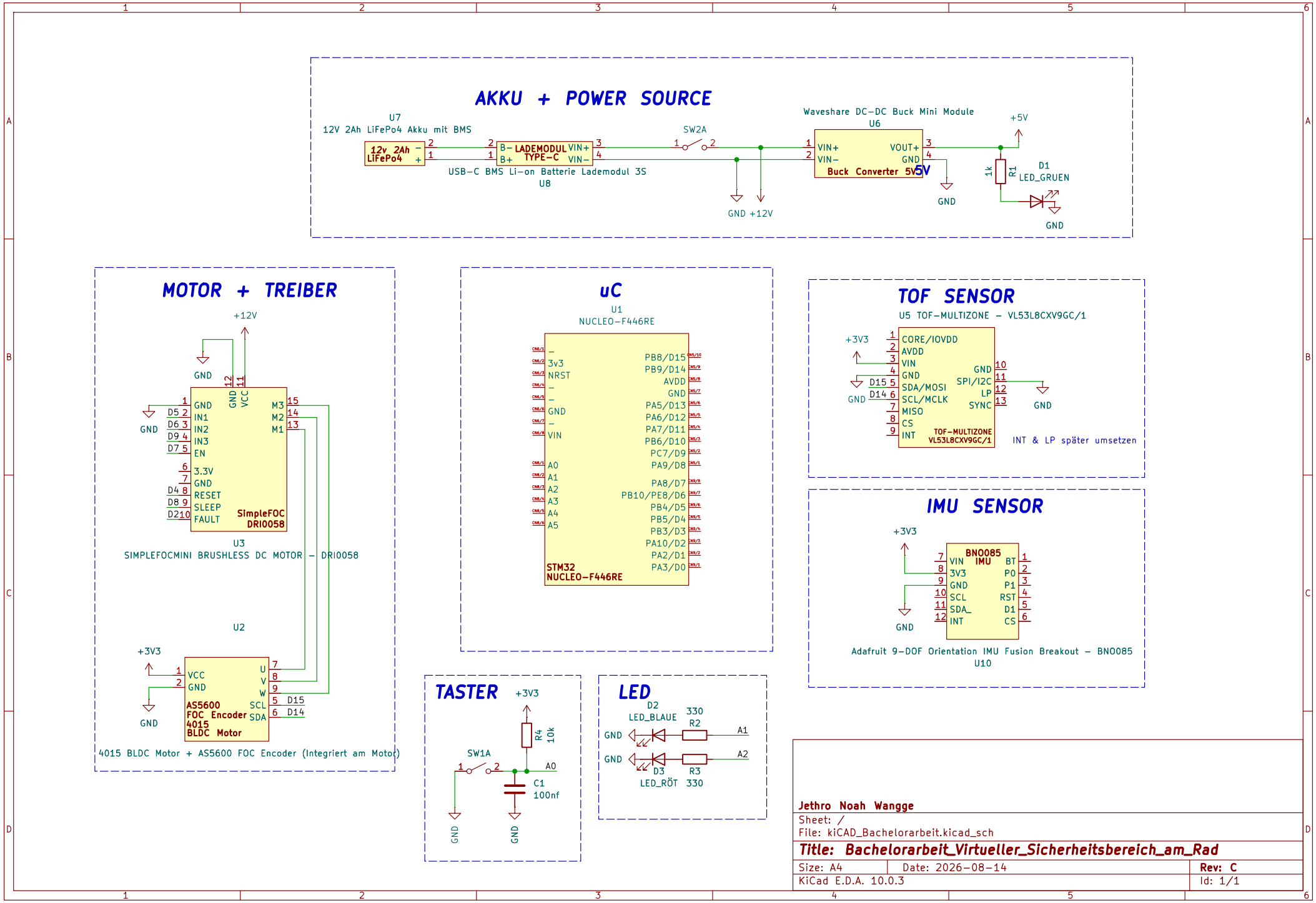

AKKU + POWER SOURCE
U7
12V 2Ah LiFePo4 Akku mit BMS
12v 2Ah LiFePo4
USB-C BMS Li-on Batterie Lademodul 3S
U8
B- LADEMODUL TYPE-C B+
VIN+ VIN-
SW2A
Waveshare DC-DC Buck Mini Module
U6
VIN+ VIN- VOUT+ GND
Buck Converter 5V
5V
GND +12V
GND
+5V
1k
R1
D1
LED_GRUEN
GND
MOTOR + TREIBER
+12V
GND
GND VCC
GND IN1 IN2 IN3 EN 3.3V GND RESET SLEEP FAULT
M3 M2 M1
D5 D6 D9 D7 D4 D8 D2
SimpleFOC DRI0058
U3
SIMPLEFOCMINI BRUSHLESS DC MOTOR - DRI0058
U2
+3V3
VCC GND
U V W SCL SDA
D15 D14
AS5600 FOC Encoder 4015 BLDC Motor
GND
4015 BLDC Motor + AS5600 FOC Encoder (Integriert am Motor)
uC
U1
NUCLEO-F446RE
3V3 NRST GND VIN A0 A1 A2 A3 A4 A5
PB8/D15 PB9/D14 AVDD GND PA5/D13 PA6/D12 PA7/D11 PB6/D10 PC7/D9 PA9/D8 PA8/D7 PB10/PE8/D6 PB4/D5 PB5/D4 PB3/D3 PA10/D2 PA2/D1 PA3/D0
STM32 NUCLEO-F446RE
TOF SENSOR
U5 TOF-MULTIZONE - VL53L8CXV9GC/1
+3V3
CORE/IOVDD AVDD VIN GND SDA/MOSI SCL/MCLK MISO CS INT
GND SPI/I2C LP SYNC
D15 D14
TOF-MULTIZONE VL53L8CXV9GC/1
INT & LP später umsetzen
IMU SENSOR
+3V3
BNO085 IMU
VIN 3V3 GND SCL SDA_ INT
BT P0 P1 RST D1 CS
Adafruit 9-DOF Orientation IMU Fusion Breakout - BNO085
U10
TASTER
+3V3
R4 10k
SW1A
A0
C1 100nf
GND
LED
D2 LED_BLAUE
330 R2
A1
D3 LED_RÖT
R3 330
A2
Jethro Noah Wangge
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